

Q1 - 24 January - Shift 1

Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A : Photodiodes are preferably operated in reverse bias condition for light intensity measurement.

Reason R : The current in the forward bias is more than the current in the reverse bias for a p - n junction diode.

In the light of the above statement, choose the correct answer from the options given below :

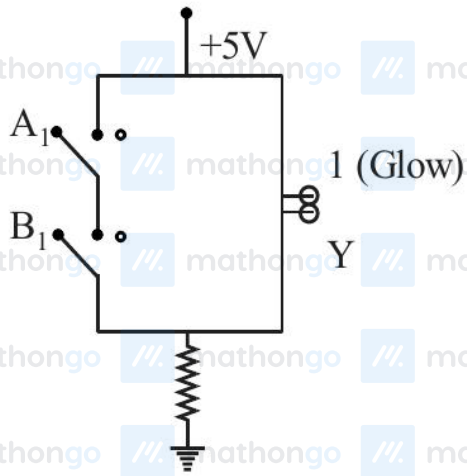
- (1) **A** is false but **R** is true
- (2) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**
- (3) **A** is true but **R** is false
- (4) Both **A** and **R** are true and **R** is the correct explanation of **A**

Space for your notes:

Q2 - 24 January - Shift 2

The logic gate equivalent to the given circuit diagram is :

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- (1) OR
- (2) NAND
- (3) NOR
- (4) AND

Q3 - 25 January - Shift 1

Given below are two statements : one is labeled as Assertion A and the other is labeled as Reason R

Assertion A: Photodiodes are used in forward bias usually for measuring the light intensity.

Reason R: For a p-n junction diode, at applied voltage V the current in the forward bias is more than the current in the reverse bias for $|V_z| > \pm V \geq |V_0|$ where V_0 is the threshold voltage and V_z is the breakdown voltage.

In the light of the above statements, choose the **correct** answer from the options given below

- (1) Both A and R are true and R is correct explanation A
- (2) Both A and R are true but R is NOT the correct explanation A
- (3) A is false but R is true
- (4) A is true but R is false

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Q4 - 25 January - Shift 2

Statement I : When a Si sample is doped with Boron, it becomes P type and when doped by Arsenic it becomes N-type semi conductor such that P-type has excess holes and N-type has excess electrons.

Statement II : When such P-type and N-type semi-conductors, are fused to make a junction, a

current will automatically flow which can be detected with an externally connected ammeter.

In the light of above statements, choose the most appropriate answer from the options given below.

Options:

- (1) Both Statement I and statement II are incorrect
- (2) Statement I is incorrect but statement II is correct
- (3) Both Statement I and statement II are correct
- (4) Statement I is correct but statement II is incorrect

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Q5 - 29 January - Shift 1

Which of the following statement is not correct in the case of light emitting diodes?

- A. It is a heavily doped p-n junction.
- B. It emits light only when it is forward biased.
- C. It emits light only when it is reverse biased.
- D. The energy of the light emitted is equal to or slightly less than the energy gap of the semiconductor used.

Choose the correct answer from the options given below :

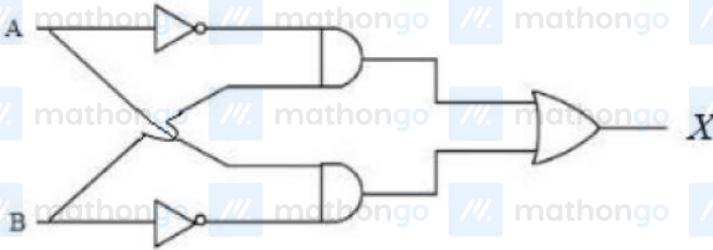
- (1) C and D
- (2) A
- (3) C
- (4) B

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Q6 - 29 January - Shift 2

For the given logic gates combination, the correct truth table will be

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(1)

A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

(2)

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

(3)

A	B	X
0	0	1
0	1	0
1	0	1
1	1	0

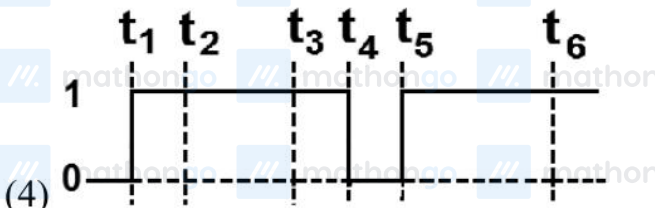
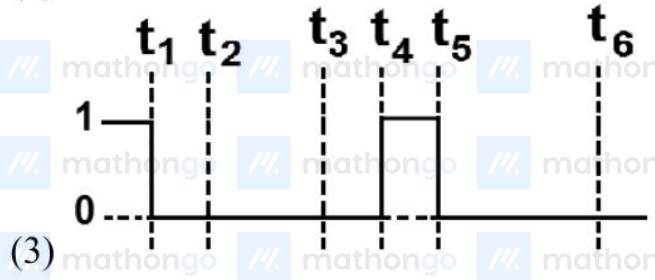
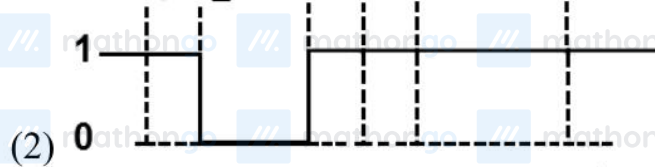
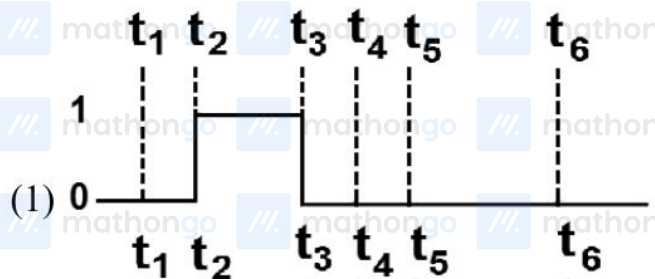
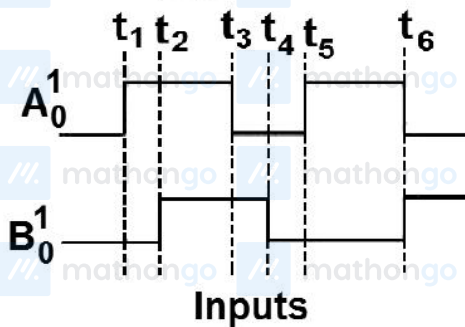
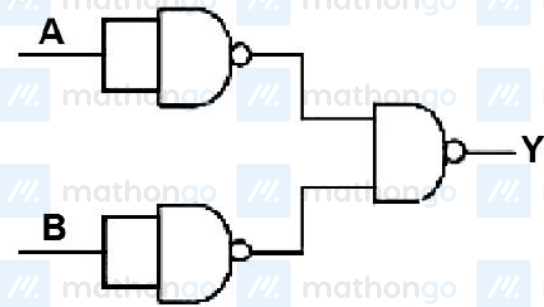
(4)

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

Q7 - 30 January - Shift 1

The output waveform of the given logical circuit for the following inputs A and B as shown below, is :

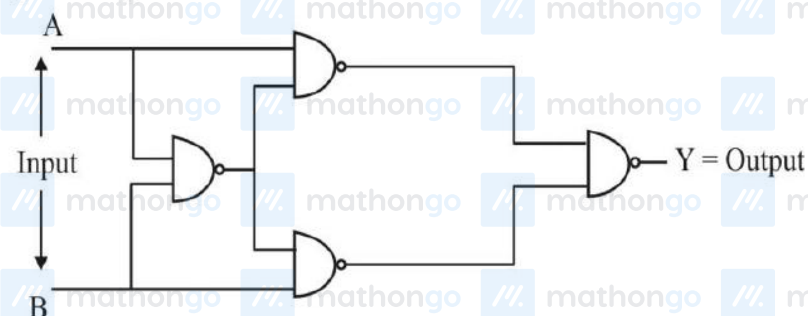
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Q8 - 30 January - Shift 2

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The output Y for the inputs A and B of circuit is given by



Truth table of the shown circuit is :

A	B	Y
0	0	1
(1) 0	1	1
1	0	1
1	1	0

A	B	Y
0	0	1
(2) 0	1	0
1	0	0
1	1	1

A	B	Y
0	0	0
(3) 0	1	1
1	0	1
1	1	1

A	B	Y
0	0	0
(4) 0	1	1
1	0	1
1	1	0

Q9 - 31 January - Shift 1

The effect of increase in temperature on the number of electrons in conduction band (n_e) and resistance of a semiconductor will be as:

- (1) Both n_e and resistance decrease
- (2) Both n_e and resistance increase
- (3) n_e increases, resistance decreases
- (4) n_e decreases, resistance increases

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Q10 - 31 January - Shift 2

Given below are two statements:

Statement I: In a typical transistor, all three regions emitter, base and collector have same doping level.

Statement II: In a transistor, collector is the thickest and base is the thinnest segment.

In the light of the above statements, choose the **most appropriate** answer from the options given below.

- (1) Both **Statement I** and **Statement II** are correct
- (2) Both **Statement I** and **Statement II** are incorrect
- (3) **Statement I** is incorrect but **Statement II** is correct
- (4) **Statement I** is correct but **Statement II** is incorrect

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Q11 - 01 February - Shift 1

Match the List I with List II

	List I		List II
A.	Intrinsic Semiconductor	I.	Fermi-level near conduction band
B.	n-type semiconductor	II.	Fermi-level at middle
C.	p-type semiconductor	III.	Fermi-level near valence band
D.	Metals	IV.	Fermi-level inside conduction band

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Choose the **correct** answer from the options given below:

(1) (A) $\rightarrow$ I, (B) $\rightarrow$ II, (C) $\rightarrow$ III, (D) $\rightarrow$ IV

(2) (A) $\rightarrow$ II, (B) $\rightarrow$ I, (C) $\rightarrow$ III, (D) $\rightarrow$ IV

(3) (A) $\rightarrow$ II, (B) $\rightarrow$ III, (C) $\rightarrow$ I, (D) $\rightarrow$ IV

(4) (A) $\rightarrow$ III, (B) $\rightarrow$ I, (C) $\rightarrow$ II, (D) $\rightarrow$ IV

Q12 - 01 February - Shift 2

Choose the correct statement about Zener diode:

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(1) It works as a voltage regulator in reverse bias and behaves like simple pn junction diode in forward bias.

(2) It works as a voltage regulator in both forward and reverse bias.

(3) It works as a voltage regulator only in forward bias.

(4) It works as a voltage regulator in forward bias and behaves like simple pn junction diode in reverse bias.

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (2)

Q2 (2)

Q3 (3)

Q4 (4)

Q5 (3)

Q6 (2)

Q7 (4)

Q8 (4)

Q9 (3)

Q10 (3)

Q11 (3)

Q12 (1)

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Q1 (2)

Photodiodes are operated in reverse bias as fractional change in current due to light is more easy to detect in reverse bias.

Q2 (2)

A_1	B_1	Y
0	0	1
0	1	1
1	0	1
1	1	0

$$Y = A_1 \odot B_1 \text{ NAND}$$

Q3 (3)

Theory based

Photodiodes are operated in reverse bias condition.

For P-N junction current in forward bias (for $V \geq V_0$) is always greater than current in reverse bias (for $V \leq V_z$).

Hence Assertion is false but Reason is true

Q4 (4)

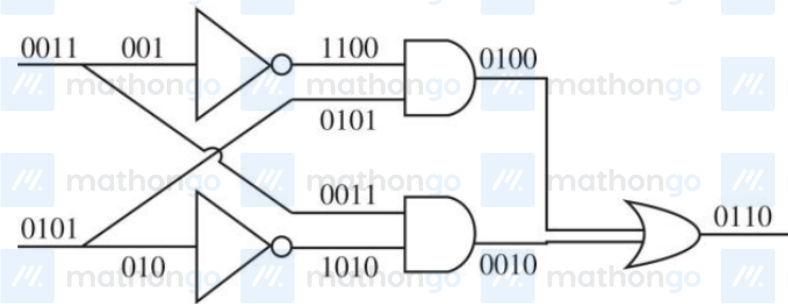
Statement – I is correct

When P-N junction is formed an electric field is generated from N-side to P-side due to which barrier potential arises & majority charge carrier can not flow through the junction due to barrier potential so current is zero unless we apply forward bias voltage.

Q5 (3)

LED works in forward biasing and light energy maybe slightly less or equal to band gap.

Q6 (2)



Q7 (4)

$$(A \cdot A) = A$$

$$B \cdot B = B$$

$$(\overline{A \cdot B}) = A + B$$

OR Gate

Q8 (4)

Given circuit represent XOR

Q9 (3)

As temperature increases, more electron excite to conduction band and hence conductivity increases, therefore resistance decreases.

Q10 (3)

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No Solution

Q11 (3)

Based on theory

Q12 (1)

Woks as voltage regulator in reverse bias and as simple P-n junction in forward bias.

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